PortfolioZero: A Stock Portfolio Model based on Deep Reinforcement Learning

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Abstract

Research on portfolio management is a prominent topic in the financial field. Existing studies primarily use reinforcement learning methods to build models aimed at achieving high investment returns while minimizing risks from market uncertainties. Two main issues exist in current research: First, the complexity of financial markets makes it challenging to capture asset price change patterns. Second, current research mainly relies on the Efficient Market Hypothesis, which assumes stock prices instantly and accurately show all asset information, suggesting that historical prices alone can predict future trends. However, in reality, numerous external factors beyond price influence future judgments. Therefore, we introduce PortfolioZero, a novel model for portfolio management. PortfolioZero is an enhanced reinforcement learning model based on the MuZero framework. It utilizes three connected deep neural networks combined with a Monte Carlo Tree to discover patterns of financial assets. In the representation network, a Transformer based pre-trained model is used to embed financial price data along the time dimension to capture temporal dynamics and potential correlations, providing richer feature representations. Moreover, the prediction network and Monte Carlo Tree Search in the model are redesigned to accommodate the continuous action space of the financial market environment. We used the StructBERT model to process financial text data, extracting market sentiment information into sentiment scores. These scores are used to reconstruct two novel reward functions to capture dynamic changes in the financial market. In experiments, we compared our model with traditional non-reinforcement learning meth-

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ods and cutting-edge deep reinforcement learning algorithms. PortfolioZero demonstrated higher returns and more robust risk-control capabilities across various market environments.

Keywords: PortfolioZero, Deep Reinforcement Learning, Portfolio, MuZero, Pre-trained Models

1. Introduction

To manage risks and optimize returns in financial markets, investors employ portfolio management strategies, which often include diversifying asset holdings. This strategy is based on long-term return goals and risk tolerance, choosing and optimizing asset selections by considering the expected return and risk parameters of each asset. Stocks, as representative assets, can be effectively managed to increase returns while reducing risks. A critical aspect of this process is portfolio rebalancing, where investors routinely adjust their portfolios to maintain predetermined strategic asset allocation ratios.

Reinforcement learning (RL) is a hot research area in artificial intelligence, where trained agents perceive environments and autonomously select optimal actions. This technology has achieved significant success in fields such as video games and Go [39] along with the advancements in computational power and deep learning (DL). RL is becoming a powerful fintech tool [41], aiding investment decisions in complex, non-linear financial markets, especially in portfolio management. The problem is how to train efficient agents capable of thoroughly analyzing financial asset characteristics, optimizing decisions to maximize returns through deep perception of historical data.

Early research in portfolio management relied on Recurrent Reinforcement Learning (RRL), followed by Deep Reinforcement Learning (DRL) approaches incorporating various Deep Neural Networks (DNN) and financially-informed reward functions. However, Existing research has two main limitations: First, they fail to fully explore potential features from historical trading data, resulting in incomplete state representation for reinforcement learning and reducing learning efficiency; second, it neglects external factors such as market sentiment and macroeconomic conditions, which are precisely the key variables influencing stock trends. In recent years, some studies have focused on these limitations. For example, Ye et al.[46] explores the integration of trading data with news data to enhance the state representation

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in reinforcement learning. However, when news data is transformed into opinion-based information, it becomes extremely sparse. If directly merged with status data, it is likely to be overwhelmed by massive transactions. If the deep network design is not comprehensive enough and hyper-parameter adjustments are insufficient, it will be difficult to achieve the desired guiding effect. Thus, how to effectively utilize sparse discrete news data still requires further research.

We present PortfolioZero, a novel reinforcement learning model based on the MuZero framework[36]. In PortfolioZero model, we try to use the new architecture and the market sentiment based reward functions to address the limitations. The specific contributions are as follows.

1. **Innovative Model Framework**: PortfolioZero introduces a novel reinforcement learning framework for portfolio management by unifying three complementary deep neural networks—representation, dynamics, and prediction—along with a Monte Carlo Tree Search (MCTS) [8]. Because the action space is continuous, we embed insights from deterministic policy gradient: the prediction network now outputs deterministic actions, while the MCTS procedure incorporates Ornstein-Uhlenbeck (OU) noise [42] to broaden the action space and enhance exploration. To the best of our knowledge, such an approach has not yet been applied in prior portfolio studies. By enabling more analysis of financial asset fluctuations and facilitating continuous rebalancing strategies, PortfolioZero effectively addresses the second challenge outlined above.
2. **Temporal Feature Extraction**: Previous works usually used the recurrent networks to abstract the features of the data. In this paper, we employ a Transformer-based pre-trained model to capture temporal dynamics and correlations in financial price data. This approach significantly improves the model to represent complex market behaviors, thus addressing the first challenge.
3. **Market Sentiment Integration**: The model incorporates market sentiment analysis through the StructBERT model, which processes financial news and commentary to extract sentiment scores. These scores are utilized to construct two novel reward functions that reflect market sentiment, thereby allowing the model to adapt its investment strategies in response to changing market conditions. This method can effectively raise the capability of our model on exploitation, and thus

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address the second challenge.

4. **Robust Performance Evaluation**: Extensive empirical evaluations demonstrate that PortfolioZero outperforms traditional non-reinforcement learning methods and state-of-the-art deep reinforcement learning algorithms across various market environments. The model consistently achieves higher returns and exhibits superior risk management capabilities, particularly in upward, downward, and volatile market conditions.

The rest of the paper is organized as following: Section 2 introduces relevant theories of financial investment and related works, and provides a formal definition of the problem addressed in this paper. Section 3 presents the PortfolioZero model with a detailed description of its overall design and optimization. Section 4 evaluates the effectiveness and performance of our model through experimental studies. Section 5 concludes the paper.

2. Related Works and Problem Statement

2.1. Financial Investment Theory

The Efficient Market Hypothesis (EMH)[12], proposed by Fama et al., is a classic theory in finance studies. According to the EMH, financial markets are highly efficient, with all available information already reflected by asset prices. Consequently, investors cannot consistently achieve excess returns by analyzing public information. However, with the development of modern technology, this hypothesis has been challenged.

Lo, incorporating evolutionary theory and the concept of bounded rationality, proposed the Adaptive Market Hypothesis (AMH) [29]. The AMH views the market as a dynamic feedback system, in which responses are not limited to simple gains and losses. Within this framework, the efficiency of financial trading systems should not be assessed solely based on single transaction outcomes but observed from a long-term perspective, such as annual or quarterly total returns, which often exhibit delayed characteristics. The AMH considers financial markets as an ever-evolving ecological environment with various market participants whose rational behaviors are not always immediate and whose impact on financial product prices is not direct. This dynamism fosters active liquidity in financial markets, implying the presence of arbitrage opportunities readily captured by market participants. The AMH emphasizes that market liquidity and arbitrage opportunities are continually changing. Under evolutionary pressures, changes in liquidity alter the trading

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environment and affect the entire business ecosystem. Therefore, an efficient financial trading system must sensitively perceive market changes and take corresponding actions, such as going long, short, or holding positions, to adapt to ever-changing market conditions.

2.2. Related Works

2.2.1. Traditional Portfolio Management

Traditional portfolio management methods can be categorized into four types: "Follow the Winner," "Follow the Loser," "Pattern Matching," and "Meta-Learning" [24]. "Follow the Winner" and "Follow the Loser" fall under momentum-based investment strategies. The "Follow the Winner" strategy focuses on investing in assets that have performed well in the past, while "Follow the Loser" targets poorly-performing assets. Both strategies typically rely on qualitative analysis of short-term market trends instead of long-term fundamental analysis, making their effectiveness susceptible to market volatility.

"Pattern Matching" involves making decisions based on identifiable patterns in historical data. In Markowitz's mean-variance theory [31], investors determine optimal asset allocation ratios by analyzing historical returns and volatilities to achieve the goals of maximizing expected returns or minimizing risk. This model introduced mathematical statistics into portfolio selection, significantly advancing modern portfolio theory.

"Meta-Learning" often relies on machine learning or artificial intelligence technologies, automatically adjusting investment decisions to respond to market changes. The effectiveness of this approach often relies on the design of the technologies.

Overall, each strategy has its unique advantages and drawbacks, providing investors a range of choices. Thus, investors need to thoroughly understand the applicability and potential limitations of each method to select the strategy that best aligns with their needs and objectives.

As financial markets exhibit characteristics of asymmetric distribution of security returns and asymmetric correlations, the mean-variance model has been increasingly questioned. A key issue is that this model may underestimate the tail risk of portfolios [1]. Particularly during significant market downwards, asset correlations often surge, potentially leading to substantial opportunity costs. Furthermore, in real world, investors face constraints such as available capital, market interest rate limitations, minimum trading units, and transaction costs, all of which restrict the flexibility of asset allocation.

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To enhance the practicality of modern portfolio methods, many scholars have proposed improvements to the mean-variance model. For example, Sharpe [38] introduced a simplified version of the mean-variance model to enhance its usability. Konno et al. [22] replaced variance with absolute deviation as a measure of risk to reflect it more directly. Jorion [21] introduced Value-at-Risk (VaR) as a new dimension for risk assessment. David et al. proposed a multiplicative update rule EG [17] for rapid portfolio rebalancing.

The rapid development of machine learning has led to its widespread application in financial trading markets, due to its ability to deeply extract hidden information from large datasets. Borodin et al. proposed the ANTICOR algorithm [6], which enhances model prediction capabilities by using prediction components from the Lempel-Ziv lossless compression algorithm combined with follow-the-leader strategies and regularization techniques to achieve optimal asset allocation in an online learning environment. In 2011, Li et al. introduced the CORN algorithm [25], effectively utilizing statistical relationships between windows in the stock market through non-parametric learning methods, allowing it to outperform market indices and top-performing stocks. Heaton et al. [16] proposed the theory of deep investing, using neural network-based objective functions to explore the deep mapping relationship between portfolios and objectives. This study demonstrated the potential and effectiveness of deep learning in asset price prediction and resource allocation. Wang et al. [44] developed a hybrid deep learning method that combines LSTM networks with the mean-variance model. This method predicts asset returns through LSTM networks, selects high-potential assets for preliminary screening, and then uses the mean-variance model for portfolio optimization. Cao et al. [7] tackled the non-convex optimization problem in the mean-variance model using dynamic neural networks, significantly reducing the cost function by 123.6%.

2.2.2. Portfolio based on DRL

The evolution of Reinforcement Learning (RL) in quantitative financial trading reveals a progressive refinement of algorithmic approaches, primarily focusing on RRL and Q-Learning Reinforcement Learning (QRL). Pioneered by Moody et al. in 1998 [32], RRL initially addressed reward sparsity in high-frequency trading by combining recurrent neural networks with reinforcement learning, demonstrating adaptability in index trading [13].

Subsequent research expanded RRL through innovative strategies: proposed adaptive portfolio management [19], risk assessment techniques [4],

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multi-strategy models [30], Gorse proposed trading cost optimization [14], genetic algorithm integration [47], and advanced trading systems [2, 3].

While RRL showed promising adaptability, its limitations in state and action representation prompted exploration of alternative RL methodologies. QRL, despite initial limitations, offered increased flexibility. Moody's early QRL applications [33], Lee's multi-agent systems [23], and Li's risk-sensitive reward functions [26] demonstrated potential improvements. Bertoluzzo [5] and Du [11] further expanded state variable definitions and action spaces, with Pendharkar implementing SARSA algorithms in retirement portfolio management [48, 35].

In 2006, Hinton et al. [18] ushered in a new era of deep learning by introducing unsupervised learning with deep belief networks. In 2017, Deng et al. [10] proposed the financial trading system FRDNN, combining fuzzy learning, DNN, and RRL to reduce data uncertainty, efficiently denoise, and extract features. Jiang et al. [20] introduced a model-free RL framework for cryptocurrency investment, using DNN to optimize deterministic policies. This framework inspired the application of various RL algorithms in portfolio management, including DDPG [27], PPO [37], and SAC [15]. Liu et al. developed a library named FinRL [28], which consolidates reinforcement learning methods used in automated stock trading in quantitative finance. In 2021, Soleymani et al. [40] employed graph convolutional neural networks to extract features from financial data, revealing inter-asset relationships.

In recent years, Ye et al. [46] proposed the SARL framework to integrate heterogeneous data, including traditional financial data and news articles, for asset price prediction within an RL model. Wang et al. [45] introduced the DeepTrader algorithm, leveraging temporal convolutional networks (TCN), attention mechanisms, and graph convolutional networks (GCN) to handle inter-asset relationships and dynamically adjust to market conditions, achieving a balance between risk and return. The introduction of new data sources in SARL and DeepTrader offer fresh perspectives and methods for portfolio research.

Notably, the proposed PortfolioZero, similar to SARL, employs news data as an additional data source and has achieved commendable results. The primary difference between the two methods lies in how they utilize news data: SARL introduces news data as state in reinforcement learning, whereas our model quantifies market sentiment to construct reward functions for reinforcement learning. This distinction leads to differences in subsequent data organization and model construction.

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2.3. Problem Statement

Let Φ denote the set of assets in a portfolio, consisting of N assets, each represented as Φ_i for $i = 1, \ldots, N$. The portfolio includes a risk-free asset, typically cash, denoted as Φ_0. At the end of trading period $t \in \{0, \cdots, T\}$, the price vector of the i-th asset is denoted as $\mathbf{v}_{i,t}$.

The relative price vector $\mathbf{y}_t$ for a portfolio with N assets at time t is defined as:

$$\mathbf{y}_t = \frac{\mathbf{v}_t}{\mathbf{v}_{t-1}} = \left(\frac{v_t^0}{v_{t-1}^0}, \frac{v_t^1}{v_{t-1}^1}, \frac{v_t^2}{v_{t-1}^2}, \ldots, \frac{v_t^N}{v_{t-1}^N} \right)^T \tag{1}$$

Here, $\frac{v_t^0}{v_{t-1}^0} = 1$.

In each trading period, the portfolio weight vector is represented as:

$$\mathbf{w}_t = \left(w_t^0, w_t^1, \ldots, w_t^N \right)^T \tag{2}$$

The $w_{i,t}$ represents the weight in the i-th asset at time t. The weights sum to 1: $\sum_i w_{i,t} = 1$.

The portfolio value p_t is calculated by the dot product of the current relative price vector $\mathbf{y}_t$ and the portfolio weight vector $\mathbf{w}_{t-1}$ of previous period is:

$$p_t = p_{t-1} \mathbf{y}_t \cdot \mathbf{w}_{t-1} \tag{3}$$

In real markets, buying and selling incur costs, like commissions. Let μ_t be the discount factor for each transaction. Jiang et al. [20] showed that this factor converges through iterative fees. Thus, accounting for costs, the portfolio value update is:

$$p_t = \mu_t p_t' = \mu_t p_{t-1} \mathbf{y}_t \cdot \mathbf{w}_{t-1} \tag{4}$$

Market price changes necessitate the use of Ormos and Urbán's [34] recursive formula to update portfolio weights. The new weights are based on previous weights $\mathbf{w}_{t-1}$ and the relative price vector $\mathbf{y}_t$:

$$\mathbf{w}_t' = \frac{\mathbf{y}_t \odot \mathbf{w}_{t-1}}{\mathbf{y}_t \cdot \mathbf{w}_{t-1}} \tag{5}$$

where $\odot$ denotes element-wise multiplication, and $\cdot$ the dot product. This formula adjusts $\mathbf{w}_t'$ to reflect the market value changes while maintaining a weight sum of 1.

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Reallocation of weights changes the value of portfolio. The reward r_t quantifies strategy performance at each step:

$$r_t = \ln\left(\frac{p_t}{p_{t-1}}\right) \tag{6}$$

The final portfolio value p_T considers the cumulative reward over the entire period, reflecting the import of strategy:

$$p_T = p_0 \exp\left(\sum r_t\right) \tag{7}$$

Propostion 1 (Problem Definition)**.** *The objective in this paper is to learn a policy π, which can provide a strategy to maximize the final portfolio value p_{T+k}= $p_T exp(\sum_{t=T}^{T+k} r_t)$ by adjusting portfolio weights $\mathbf{w}_t$ at each time step t.*

3. PortfolioZero Model

This section presents the design, implementation, and optimization of the PortfolioZero model. By integrating DRL with AMH theory, we offer a novel perspective on market dynamics, providing new solutions for stock market portfolio management.

3.1. States, Actions and Rewards

3.1.1. States

At time t, current market data is observed to provide context for sequential decision-making. Considering its correlation with the previous moment, the observation vector $\mathbf{o}_{t-1} \in \mathbb{R}^{N \times M_{t-1} \times F}$, where N is the stock number, M the length of the sliding window at t, reflecting historical price influence, and F is the dimension of features for each stock, usually including historical data like high, low, and closing prices.

The state is a latent vector derived from a representation network (Section 3.3.1), not the observation vector. Thus, the state space consists of latent vectors, with dimensions defined by the output of representation network.

3.1.2. Actions

For each state $\mathbf{s}_t$, the agent determines an action $\mathbf{a}_t \in \mathbb{R}^N$, a weight vector for the assets. This vector includes the allocation of both risk-free and risky assets, formally represented as:

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$$\mathbf{a}_t = (w_t^0, w_t^1, \ldots, w_t^N) \quad 0 \leq w_t^i \leq 1 \; \forall i \in \{0, \ldots, N\} \tag{8}$$

Each weight can be any real number between 0 and 1, making the action space continuous. Generating actions is the main task, which will reallocate assets at each time point.

3.1.3. Basic Reward Function

In financial investment, the goal is to maximize returns. Therefore, a fundamental reward function is constructed based on the average logarithmic return over the entire investment period:

$$R(s_1, a_1, \ldots, s_T, a_T) = \frac{1}{T} \sum_{t=1}^{T} r_t \tag{9}$$

By maximizing reward function R, the agent learns to make decisions and maximize portfolio returns over the investment period.

3.2. Reward with Market Sentiment

Due to the highly dynamic and nonlinear nature of the financial market, it is not enough to only rely on price information for investment decisions. Previous studies indicate that market sentiment significantly impacts investment decisions. To verify this hypothesis, this study compares market sentiment expressions with actual market data.

We randomly selected stock as the observation subject and monitored from March 1, 2019, to March 1, 2022, covering 727 trading days. Stock sentiment was quantified by analyzing news reports and comments related to the stock. Sentiment scores are divided into positive and negative components, with their sum always equaling 1 in each sample. The actual market performance is measured by daily stock returns.

Figure 1 illustrates the relationship between sentiment scores and price changes of the observed stock over a selected period. In the figure, part (a) shows the trend between positive sentiment scores and price changes, while part (b) highlights cases of directional alignment, which occurs when a positive sentiment score over 0.5 coincides with a positive price change, or a score below 0.5 aligns with a negative price change. We can see that 80.3% of trading days exhibit directional alignment between sentiment scores and price changes, indicating a strong correlation with market trends.

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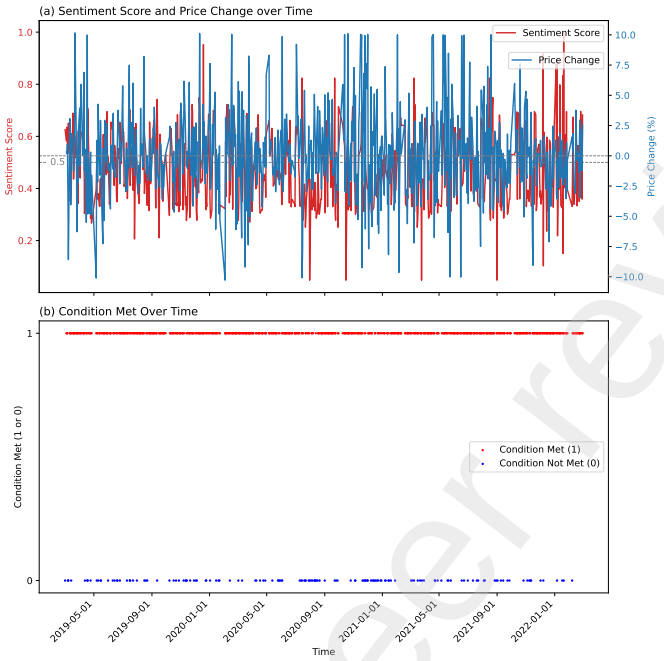


Figure 1: Sentiment Score VS Stock Price Change

Consequently, we propose two sentiment-based reward functions to integrate market sentiment into reward design. The goal is to reward decisions aligned with market sentiment, guiding the DRL algorithm to adjust portfolio allocations under varying market conditions. The basic idea is that if the predicted investment decision aligns with sentiment-driven market trends, it receives a positive reward. Conversely, if a decision contradicts the sentiment trend, a negative reward is given. This mechanism enhances the sensitivity of model to the market sentiment.

The first reward function, R^{α}, is a weighted sum of each asset based on sentiment scores:

$$R^{\alpha} = \sum_{i=1}^{N} w_i \times \text{scores}_i \tag{10}$$

where w_i is the weight of the i-th asset in the portfolio, and scores_i is the sentiment score of that asset. This method multiplies each sentiment score by its weight, linking sentiment value with market performance. Intuitively, it amplifies sentiment scores on asset weights when aligned.

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Table 1: An Example of Three Stocks

t		i=1	i=2	i=3
1	score	0.3	0.3	0.4
	weight	0.3	0.2	0.5
	rank	2	1	3
2	score	0.1	0.4	0.5
	weight	0.2	0.5	0.3
	rank	2	3	1
3	score	0.3	0.1	0.6
	weight	0.2	0.1	0.6
	rank	2	1	3

The second reward function, R^β, considers the relative change in asset weights and sentiment score magnitudes:

$$R^\beta = \sum_{i=1}^{N} \sum_{j=1}^{N} (\text{scores}_i - \text{scores}_j)(\text{rank}_i - \text{rank}_j) \tag{11}$$

Here, rank_i represents the ranking of changes in asset weight, determined by arranging weight changes from period $t-1$ to t in ascending order for each asset i:

$$\text{rank}_i = \text{index}(i, \text{sorted}(\{w_{i,t} - w_{i,t-1} \mid i = 0..N\})) \tag{12}$$

This approach ensures that higher sentiment scores lead to greater weights, while lower scores lead to reduced weights, optimizing asset allocation by encouraging consistency with expected performance.

The R^α is linear, evaluating the independent performance of each asset. In contrast, R^β uses a nonlinear approach with interaction terms, considering relative performance differences between assets. Additionally, R^α produces only positive values, which mainly differentiate action quality by comparing rewards from various actions. Conversely, R^β provides both positive and negative values, offering direct feedback on beneficial and adverse actions. This feedback accelerates the learning process by identifying and minimizing adverse actions. We will demonstrate these differences with an example.

Example 1: Given the portfolio of three assets shown in Table 1, where t represents time and all weights are zero at $t = 0$. Here i is the asset index.

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Calculations yield $R^{\alpha} = (0.35, 0.37, 0.43)$ and $R^{\beta} = (0.3, -0.3, 1.5)$, representing rewards for the three time points, respectively.

Both reward functions reflect the consistency between sentiment scores and asset weights; greater consistency results in higher reward scores. The distinction lies in their evaluation methods. At $t = 1$ and $t = 2$, R^{α} and R^{β} scores are inversely related.

We observed that R^{α} reflects the current consistency between sentiment scores and asset weights, whereas R^{β} captures the alignment between sentiment scores and changes in asset weights over time.This approach aligns more closely with investor perceptions, considering both the current allocation and adjustments made in response to changing market sentiments.

3.3. Model Design

We have developed PortfolioZero, an improved model based on MuZero, designed to learn the dynamic characteristics of the environment through interaction. PortfolioZero consists of three modules implemented using deep neural networks: the representation, dynamics, and prediction networks. Additionally, Monte Carlo Tree Search is utilized for network optimization.

3.3.1. Representation Network

The time-series nature of financial data means we need to incorporate more historical data into our modeling to uncover additional temporal features. To address this issue, the representation network h converts environmental observation vector o_t into a hidden state s_t, denoted as $s_t = h(o_t)$. This transformation creates a compressed informative state representation, reducing data dimension while preserving critical information.

The dynamics network presented later in the model requires long-range gradient back propagation, which will potentially lead to the vanishing gradient problem, affecting learning efficiency and model expressiveness. To address this problem, we use a pre-trained Transformer-based model for the representation network. This method leverages the self-attention mechanism and residual connections to facilitate smooth gradient propagation in deeper layers, effectively avoiding the vanishing gradient issue. Moreover, this architecture enables the transfer of rich knowledge from other data to the current task, which can significantly enhance state representation capabilities.

During pre-training process, the model updates the Transformer encoder and decoder parameters through two tasks, that is, to predict the price of next day and the next five days. In this phase, the encoder input is $o_{\text{enc}} \in$

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$\mathbb{R}^{N\times\ M\times F}$ and the decoder input is $o_{\text{dec}} \in \mathbb{R}^{N\times\ D\times F}$, where $D = 1$ or 5, indicating the observed price features over the next D days. The output of the encoder is the hidden state $s_{\text{enc}} \in \mathbb{R}^{N\times\ M\times H}$, where H denotes the dimension of the hidden state. The loss function uses the mean squared error (MSE), shown as follows. Here y_i^d and $\widehat{y}_i^d$ are the real and predicted price of next d days in sample i.

$$\mathcal{L}_D^P(\phi) = \frac{1}{D \times N} \sum_{i=1}^{N} \sum_{d=1}^{D} (y_i^d - \widehat{y}_i^d(\phi))^2 \tag{13}$$

In the subsequent reinforcement learning, only the encoder processes observation vectors to output the hidden state. Although the encoder processes input data and generates output during this phase, its gradients do not optimize the loss function of pre-training phase. Instead, the output of encoder feeds into other networks, whose loss functions relate directly to subsequent task objectives. Thus, the gradients of encoder will indirectly impact other modules.

3.3.2. Prediction Network

In a portfolio management environment with a continuous action space, we use an DDPG-alike method to construct the prediction network f. The prediction network f takes the hidden state s_t as input. Its output includes the next action a_t and the state value v_t, represented as $(a_t, v_t) = f(s_t)$. Here, "action" refers to the asset weights for the next investment period, while "value" assesses the current state.

To ensure rational asset weights, the prediction network applies a softmax function to standardize the output vector. This ensures each asset weight $w_{i,t}$ is between 0 and 1, with their sum equaling 1. Additionally, it also outputs a scalar through two fully connected layers, providing a current state evaluation score.

The prediction network serves two main purposes. The first is to leverage the current state to predict subsequent actions and evaluate the current state. It also guides action selection during sequential decision-making simulations by Monte Carlo Tree, ultimately determining a more accurate action $\hat{a}_t$ for execution.

3.3.3. Dynamics Network

Once the prediction network provides the action a_t, the dynamics network g uses the current hidden state s_t and action a_t to predict the next hidden

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state s_{t+1} and immediate reward r_t. This is expressed as $(r_t, s_{t+1}) = g(s_t, a_t)$.

The dynamics network acts as an internal simulator of interaction between the agent and the environment, enabling the algorithm to consider potential future states without directly accessing the actual environment. Although the true hidden state cannot be directly obtained, gradient propagation from the prediction network training ensures the predicted s_{t+1} approximate the true hidden state $\hat{s}_{t+1}$.

Additionally, we train the dynamics network to ensure the predicted reward r_t closely approximates the true reward $\hat{r}_t$. This true reward will combine both current returns and market sentiment scores.

3.3.4. Monte Carlo Tree Search

We used Monte Carlo Tree Search (MCTS) in our model to simulate and identify optimal actions, providing optimization targets for the prediction network. This enables the prediction network to refine its parameters and update the parameters of other two networks simultaneously. Traditional MCTS is mainly used in handling discrete action spaces. We improved MCTS to adapt it for the continuous action space in its expansion step. After the prediction network computes a deterministic action a_t and state value v_t, we introduced OU noise to a_t, adding random perturbations, and generate $\iota - 1$ additional action sets with different OU noise, forming the action set $a_{t,1}, \ldots, a_{t,\iota}$. The dynamics network uses these actions to generate ι child nodes, representing ι new state. After multiple layers of selection, expansion, and back propagation, the most frequently chosen action among the ι actions, $\hat{a}_t$, is used to train the prediction network. Note that the simulation of MCTS is only used in training process rather in policy execution, since we assume the training has optimized the strategy enough so that the execution efficiency can be enhanced. Thus, $\hat{a}_t$ serves as the optimization target for the predicted action a_t by the prediction network. To prevent excessive expansion of the search space, we set ι to 4 in this study.

3.3.5. Model Training and Optimization

The model training process is shown in Algorithm 1. We trained the three networks in the PortfolioZero model by randomly selecting K 3-tuples, that is, $(o_t^0, a_t^0, \hat{r}_t^0, \cdots, o_t^{K-1}, a_t^{K-1}, \hat{r}_t^{K-1})$. The training process begins by inputting observation o_t into the representation network to obtain the state s_t. The prediction network provides the predicted action a_t and value v_t, which are refined into the target action $\hat{a}_t$ using MCTS. State s_t and action $\hat{a}_t$ are

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Algorithm 1 PortfolioZero Training

Require: Num of steps K, Epoch E, Episode T, Discount factor γ, Learning rate η;

Initialize parameters θ;
Initialize replay buffer $\mathcal{B}$;
for $e = 1$ to E **do**
  Execution Phase:
  Reset environment, receive initial observation o_1;
  $s_1 \leftarrow h_\theta(o_1)$;
  for $t = 1$ to T **do**
    $(a_t, v_t) \leftarrow MCTS(s_t)$;
    $r_t, s_{t+1} \leftarrow g_\theta(s_t, a_t)$;
    Execute action a_t, observe $o_{t+1}, \hat{r}_t$;
  end for
  Store Trace $(o_1, a_1, \hat{r}_1, \cdots, o_T, a_T, \hat{r}_T)$ into $\mathcal{B}$;
  Training Phase:
  for $t = 1$ to $T - K + 1$ **do**
    Initialize $\mathcal{L}_\theta^a$, $\mathcal{L}_\theta^v$ and $\mathcal{L}_\theta^r$;
    for $k = 0$ to $K - 1$ **do**
      Get transition $(o_t^k, a_t^k, \hat{r}_t^k)$ from $\mathcal{B}$;
      $s_t^k \leftarrow h_\theta(o_t^k)$;
      $a_t^k, v_t^k \leftarrow f_\theta(s_t^k)$;
      $\hat{a}_t^k \leftarrow MCTS(s_t^k)$;
      $\hat{v}_t^k \leftarrow \sum_{i=t+k}^{T} \gamma^{i-t-k} \hat{r}_i$;
      $r_t^k, s_{t+1}^k \leftarrow g_\theta(s_t^k, \hat{a}_t^k)$;
      $\mathcal{L}_\theta^a \leftarrow \mathcal{L}_\theta^a + \mathrm{CE}(a_t^k, \hat{a}_t^k)$;
      $\mathcal{L}_\theta^v \leftarrow \mathcal{L}_\theta^v + \mathrm{MSE}(v_t^k, \hat{v}_t^k)$;
      $\mathcal{L}_\theta^r \leftarrow \mathcal{L}_\theta^r + \mathrm{MSE}(r_t^k, \hat{r}_t^k)$;
    end for
    $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_\theta^a + \mathcal{L}_\theta^v + \mathcal{L}_\theta^r + C\|\theta\|^2)$;
  end for
end for

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input into the dynamics network to predict the next state s_{t+1} and reward r_t. Note that since the true hidden state is inaccessible, it is necessary to update the representation network and dynamic network by gradient propagation from prediction network.

The primary objective of the model is to reduce the error between the predicted action a_t^k and the actual action $\hat{a}_t^k$, emphasizing consistency between predictions and actual decisions, i.e., $a_t^k \to \hat{a}_t^k$. Secondly, the model seeks to minimize the error between the predicted immediate reward r_t^k and the actually observed immediate reward $\hat{r}_t^k$, i.e., $r_t^k \to \hat{r}_t^k$. Finally, the model aims to minimize the error between the predicted state value and the target state value, i.e., $v_t^k \to \hat{v}_t^k$; here, the target state value can be obtained as a expected discounted returns from interacting with the real investment environment, expressed as $\hat{v}_t^k = \mathbb{E}[r_{t+k} + \gamma r_{t+k+1} + \cdots | o_1, a_1, \ldots, o_{t+k}, a_{t+k}]$, using the discount returns in the current episode in practice.

All model parameters are trained jointly by calculating the loss functions for policy, value, and reward prediction. Specifically, $\mathcal{L}_\theta^a$ uses cross-entropy (CE), while $\mathcal{L}_\theta^v$ and $\mathcal{L}_\theta^r$ MSE. Thus, the final loss function $\mathcal{L}_t(\theta)$ is optimized with respect to the parameters θ, where $\|\theta\|^2$ is a regularization term for the model parameters.

$$\mathcal{L}_t(\theta) = \sum_{k=1}^{K} \big(\mathcal{L}^a(a_t^k, \hat{a}_t^k; \theta) + \mathcal{L}^v(v_t^k, \hat{v}_t^k; \theta) + \mathcal{L}^r(r_t^k, r_t^k; \theta) \big) + C\|\theta\|^2 \tag{14}$$

4. Experimental Studies

This section will empirically validate the PortfolioZero model. We selected three distinct types of portfolios to represent different investor choices in the financial market, each consisting of nine stocks and a risk-free asset.

4.1. Experiment Design

4.1.1. Simulated Trading Conditions

The experiments were performed based on the simulated environment generated in FinRL [28]. To simulate the complex operations of the financial market effectively and enhance the practical applicability of our experimental model, Several simplified assumptions were proposed.

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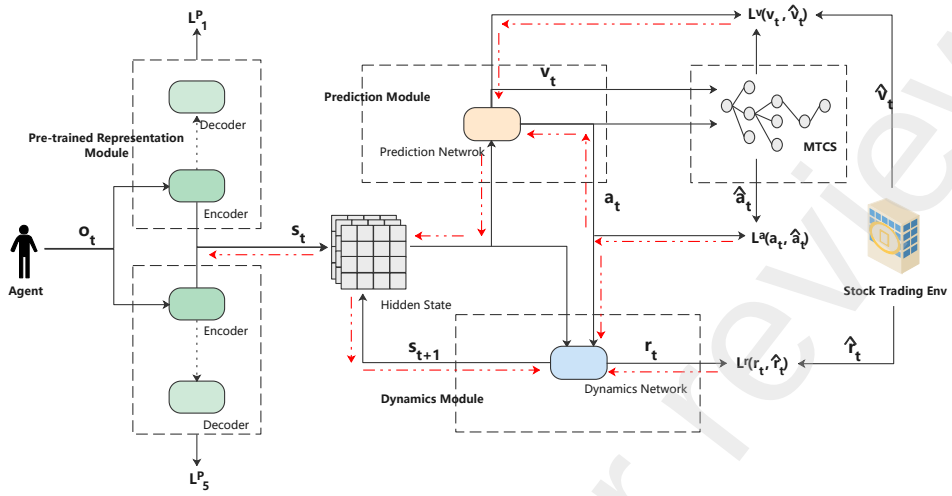


Figure 2: PortfolioZero Model: The model consists of three modules (representation network, prediction network, and dynamics network) and a Monte Carlo Tree. The black line illustrates the interaction process between the agent and the environment through the model. The red line highlights the model training process: (1) The agent observes a real state, it inputs this state into the representation network; (2) The output of the representation network is input into the prediction network and the dynamics network; (3) The prediction network generates actions and evaluates the current state; (4) The dynamics network performs model predictive control by transforming the current hidden state into the next hidden state and obtaining the immediate reward. The loss function $\mathcal{L}^a$ and $\mathcal{L}^v$ aim to minimize the error between the target and actual values and the discrepancy between predicted and actual actions, respectively. The loss function $\mathcal{L}^r$ minimizes the error between the predicted immediate reward and the observed immediate reward.

Market Impact Assumption: We assume the scale of investment funds is small enough not to significantly affect market prices.

Liquidity Assumption: We assume all assets have sufficient liquidity to ensure transactions are completed at prevailing prices.

Portfolio Adjustment Assumption: The agent reallocates portfolio weights at each predefined trading moment, with weights remaining unchanged between these moments.

Stock Supply Assumption: We assume an ample market supply of candidate investment stocks, ensuring no purchasing restrictions during operations.

In addition, to account for transaction costs during each portfolio weight adjustment, we set a uniform transaction fee rate of $\mu = 0.0025$.

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4.1.2. Data Acquisition and Preprocessing

We selected Exchange Traded Fund (ETF) from the Chinese A-share market as the experimental subjects. The daily trading data is obtained from the RESSET financial research database, including key indicators such as opening, closing, lowest, and highest prices from 2019 to 2023. Due to the characteristics of the A-share market and regulatory environment, events such as trading suspensions may lead to missing values in the dataset. To reduce the impact of these missing values on model performance, we replaced missing values with corresponding indicators from the previous trading day; plus, data standardization uses the min-max standardization method.

To simulate the asset allocation of different investors in the financial market comprehensively and evaluate the robustness of model, we constructed three portfolios using a rolling window approach to avoid look-ahead bias. The trend classification was determined using the following criteria: 1) Upward Trend: Stocks with positive 120-day rolling beta (>0.8) relative to market index. 2) Downward Trend: Stocks with negative 120-day rolling beta (<-0.8); 3) Volatile Trend: Stocks with beta between -0.8 and 0.8. Appendix Appendix A presents the criteria of stock selection and shows the details of stocks in Table A.5. We selected the upward trend portfolio to represent investors seeking long-term growth, as it consists of stocks that consistently performed well in the financial market, showing an overall price increase and ending higher than at the beginning of the backtesting period. Similarly, the downward trend portfolio contains stocks performing poorly, with overall price decreases and ending lower than at the start of the backtesting period. The volatile trend portfolio is a mixed portfolio, including six stocks with upward trends and three with downward trends.

Note the Reinforcement Learning paradigm typically excels in closed and stationary environments, where training and testing distributions maintain strong consistency. However, in the domain of portfolio optimization, models must confront dynamic and uncertain financial markets, which inherently presents an out-of-distribution (OOD) generalization challenge[9]. To evaluate model performance authentically and mitigate the distribution shift problem, we implemented a rigorous time series splitting strategy: dividing the dataset into a training set (in-sample period: 2019-03-01 to 2022-03-01, 727 days) and an independent test set (out-of-sample period: 2023-01-03 to 2023-12-29, 242 days). This temporal splitting methodology not only ensures temporal forward-looking evaluation but also effectively prevents overfitting

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issues, thereby more accurately measuring the generalization capability of the model in real market conditions.

Additionally, we collected news and commentary data on the stocks from the Guba website (http://guba.eastmoney.com) to extract market sentiments, which align with these periods to the trading data. Some relevance filter rules were considered: We mainly selected the news direct mentions of target stocks or companies, the industry-specific news affecting the selected assets, and the macroeconomic news impacting the market sector. Due to the complexity of information sources, we did not extensively consider factors such as source credibility, but rather treated all information as equal reflections of market sentiment. It is worth to note that we get the news and comments before the trading decision to avoid data leakage.

We used the pre-trained model StructBERT [43] to quantify market sentiment from news and comments. For each text data, StructBERT outputs two core metrics, i.e., a positive sentiment score and a negative sentiment score. The sum of these two scores equals 1. Since the output of StructBERT is meticulously preprocessed, these sentiment scores can be directly used in subsequent analyses without further normalization or transformation.

4.1.3. Evaluation of Models

Based on Section 3.2, the log return reward for the portfolio (Equation 9) is combined with sentiment score-based rewards (Equations 10 and 11) to derive three different reward functions: $R1 = R$, $R2 = R + R^{\alpha}$, and $R3 = R + R^{\beta}$. The corresponding models are denoted as PortfolioZero_R1, PortfolioZero_R2, and PortfolioZero_R3, respectively. To verify the performance of PortfolioZero, we compared it to the traditional deep learning algorithm DDPG, with models denoted as DDPG_R1, DDPG_R2, and DDPG_R3. We also employ the state-of-the-art model SARL [46] as the evaluation method, in which the sentiment scores are combined with the stock prices to form the states.

Additionally, we employed four non-reinforcement learning algorithms based on capital growth theory for comparison: the Equal Weight strategy, the ANTICOR algorithm [6], the EG algorithm [17], and the CORN algorithm [25]. These algorithms represent fundamental approaches in portfolio optimization that established key principles still relevant today. In addition, these algorithms serve as robust benchmarks that have withstood the test of time, consistently performing well across various market conditions and remaining widely used in industry applications. Comparing our model against

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these established methods helps demonstrate how our models can improve upon classical portfolio optimization principles.

4.1.4. Evaluation Metrics

To thoroughly evaluate the performance of the model, we used a set of quantitative evaluation metrics in the experiments.

Logarithmic Accumulation of Portfolio Value (LAPV): It directly reflects the success of portfolio management over time. An increase in LAPV signifies an increase in asset value over a certain period, with higher LAPV values indicating superior investment outcomes.

$$\mathrm{LAPV} = \log\left(\frac{p_T}{p_0}\right) \tag{15}$$

Annualized Rate of Return (ARR): It measures the average annual return on investment and is often used to compare different investments or portfolios. A higher ARR implies that a strategy yielded higher returns when annualized.

$$\mathrm{ARR} = \left(e^{\mathrm{LAPV}}\right)^{\frac{1}{T}} - 1 \tag{16}$$

Sharpe Ratio (SR): It assesses investment returns relative to risk by dividing excess returns by their volatility. A higher SR indicates that the strategy delivers higher returns per unit of risk.

$$\mathrm{SR} = \frac{\mathrm{ARR} - R_f}{\sigma_{\mathrm{ARR}}} \tag{17}$$

where R_f is the risk-free rate, and σ_{ARR} is the standard deviation of the return of portfolio, representing its volatility.

Maximum Drawdown (MDD): It measures the greatest potential decline in value an investment portfolio could suffer within a selected time frame, representing the maximum loss from peak to trough. Lower MDD values indicate effective risk management.

$$\mathrm{MDD} = \max_{\tau > t}\left(\frac{p_t - p_\tau}{p_t}\right) \tag{18}$$

where p_t represents the peak and p_τ represents the trough.

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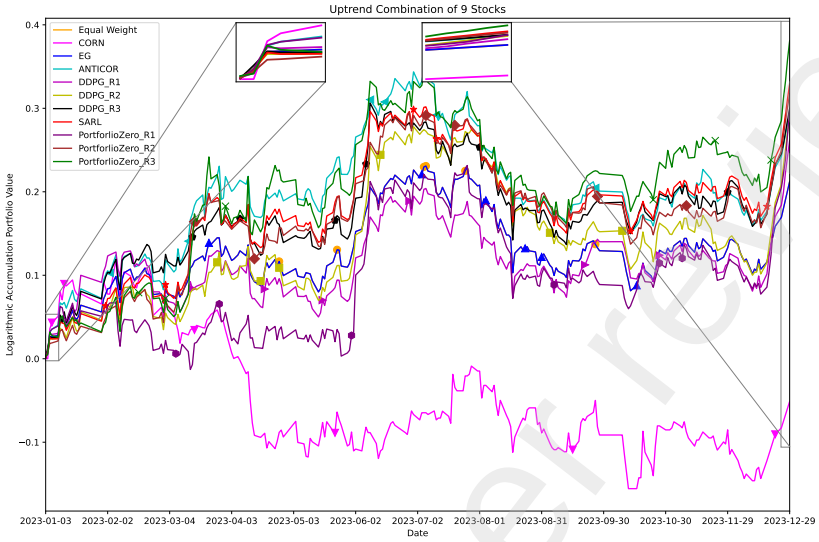


Figure 3: LAPV of Upward Trend Portfolio

Calmar Ratio (CR): It evaluates the relationship between the return of a portfolio and the risk it assumes, with a particular focus on downside risk. A high CR indicates that the strategy has achieved sustained and stable capital appreciation over the investment period.

$$\mathrm{CR} = \frac{\mathrm{ARR}}{\mathrm{MDD}} \tag{19}$$

4.2. Evaluation of PortfolioZero

4.2.1. Upward Trend Portfolio

The LAPV over time for the upward trend portfolio is shown in Figure 3. During the slight stock market rose in March 2023, other strategies showed only slight growth trends. In contrast, the PortfolioZero_R3 strategy not only achieved significant growth but also maintained a high investment return rate in the subsequent months. When the stock market experienced a slight downturn in August, the PortfolioZero_R3 strategy demonstrated strong risk control capability, keeping the LAPV curve stable without significant losses and maintaining the highest level in subsequent stages. In comparison, the CORN strategy, despite performing well in the early period, experienced a

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cliff-like decline in mid-March, which led to continued poor performance in the subsequent months, ultimately resulting in the worst performance. The ANTICOR strategy performed steadily, even surpassing the PortfolioZero_R3 before August. SARL maintained a consistent LAPV with PortfolioZero, demonstrating its ability to extract information from additional news. However, its performance was limited by using a common network, which may not fully explore the deep features of prices. When comparing the DDPG and PortfolioZero strategies, PortfolioZero generally outperformed DDPG, regardless of the reward function used. Furthermore, fixing the reinforcement learning model to compare different reward functions, performance is best with reward function R3, followed by R2, and R1 as the worst, for both DDPG and PortfolioZero. This confirms that improvements in the design of the reward function, whether through the introduction of multiple data sources or optimizations in its form, have had a positive effect.

Table 2: The Backtesting Performance of Different Strategies for the Upward Trend Portfolio

Strategy	**LAPV**	**SR**	**MDD**	**ARR(%)**	**CR(%)**
Equal Weight	0.2124	1.3785	0.1388	23.6634	170.4334
CORN	-0.0512	-0.1813	0.2395	-4.9902	-20.8393
EG	0.2124	1.3782	0.1389	23.6609	170.3859
ANTICOR	0.2952	1.5819	0.1791	34.3419	191.7730
DDPG_R1	0.2618	1.5379	**0.1222**	29.9228	244.7774
DDPG_R2	0.2896	1.6175	0.1630	33.5881	206.0550
DDPG_R3	0.3000	1.6603	0.1561	34.9827	224.1487
SARL	0.3291	**1.8614**	0.1405	38.9651	277.3867
PortfolioZero_R1	0.2988	1.4693	0.1577	34.8304	220.8810
PortfolioZero_R2	0.3176	1.8302	0.1462	37.3835	255.6218
PortfolioZero_R3	**0.3810**	1.7395	0.1629	**46.3808**	**284.8010**

Table 2 provides detailed data for different strategies in the stock market simulation. Among them, the PortfolioZero_R3 strategy standed out in LAPV performance, reaching 0.381, significantly higher than the other strategies. Regarding the SR, however, SARL led with a value of 1.8614, showing that SARL can better control the risk. Concerning the MDD, DDPG_R1 displayed a relatively low risk level with a MDD of 0.5354, showing the strategy is conservative. Although PortfolioZero_R3 did not perform as well as SARL and DDPG_R1 in terms of the SR and MDD, its more aggressive market sen-

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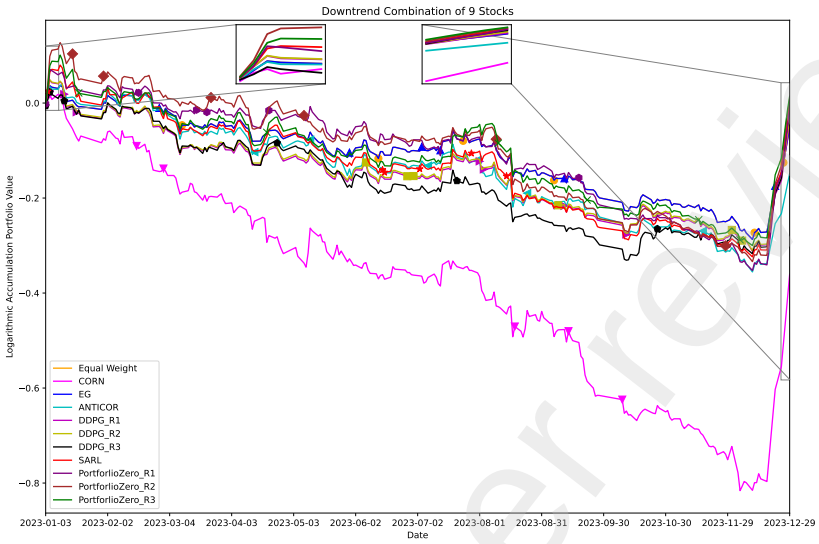


Figure 4: LAPV of Downward Trend Portfolio

timent response strategy achieved an ARR of 46.3808%, significantly higher than all other strategies. This effectively reflects the higher sensitivity of PortfolioZero_R3 to overall economic trends, favoring more risky but higher-return strategies in a generally favorable asset portfolio. Notably, due to excellent performance in ARR, PortfolioZero_R3 achieved the highest CR of 284.801%, indicating it yields higher returns when bearing the same risk, which can be considered an excellent investment strategy in practice. Among non-reinforcement learning algorithms, the ANTICOR algorithm performed best, but there remained a noticeable gap compared to reinforcement learning algorithms. The CORN strategy, due to its negative LAPV and SR, reflected higher risk and unsatisfactory returns. Comparing the two reinforcement learning algorithms, PortfolioZero generally outperformed DDPG. Furthermore, when fixing the reinforcement learning model, the effect is best with reward function R3, followed by R2. Again, this indicates that the model and reward functions used in this paper are highly effective. Note that although SARL only ranked first in SR, its other metrics were not far behind, indicating that SARL had a certain advantage in its algorithm.

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4.2.2. Downward Trend Portfolio

Figure 4 shows the LAPV of different strategies in a downward trend portfolio. The figure illustrates that, given the experimental data exhibits a downward trend, most strategies maintained negative LAPV throughout the backtesting period and performed worse than the equal-weight strategy, indicating the portfolio faced losses. Nevertheless, the PortfolioZero strategy experienced the least loss and even achieved a slight profit in the end, demonstrating its ability to identify negative features from market sentiment and effectively execute stop-loss operations. Of course, we can not deny that the profits of PortfolioZero mainly concentrated during the overall market uptrend. When market was downward, it even lagged behind non-reinforcement learning algorithms like EG. Unlike the upward trend portfolio, in the downward trend, strategies using the reward function R2 typically performed better than those using R3, which shows that R2 is also useful for special conditions.

Table 3: The Backtesting Performance of Different Strategies for the Downward Trend Portfolio

Strategy	**LAPV**	**SR**	**MDD**	**ARR(%)**	**CR(%)**
Equal Weight	-0.0527	-0.3040	0.2769	-5.1289	-18.5231
CORN	-0.3607	-1.2457	0.5658	-30.2800	-53.5161
EG	-0.0527	-0.3042	**0.2769**	-5.1306	-18.5291
ANTICOR	-0.1485	-0.8288	0.3256	-13.7987	-42.3753
DDPG_R1	-0.0430	-0.2149	0.3025	-4.2122	-13.9225
DDPG_R2	-0.0322	-0.1419	0.3022	-3.1669	-10.4808
DDPG_R3	-0.0161	-0.0328	0.2980	-1.5953	-5.3527
SARL	0.0022	0.0597	0.3322	0.2237	0.6734
PortfolioZero_R1	-0.0053	0.0246	0.3448	-0.5276	-1.5300
PortfolioZero_R2	0.0093	0.0916	0.3693	0.9373	2.5379
PortfolioZero_R3	**0.0144**	**0.1190**	0.3304	**1.4475**	**4.3813**

Table 3 details the performance of each strategy using the evaluation metrics for this portfolio. The agent utilizing the PortfolioZero_R3 strategy performed the best across four core evaluation metrics. Despite the overall market downturn, this strategy maintained the LAPV at 0.0144, demonstrating its capability to achieve certain returns even in a declining market. Additionally, it achieved an ARR of 1.4475% and a CR of 4.3813%. The lowest MDD in this experimental group was achieved by the EG and Equal

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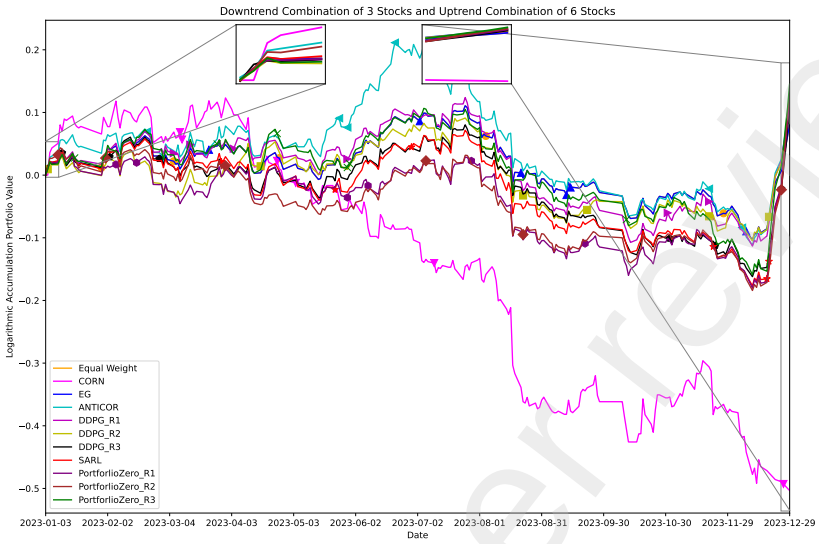


Figure 5: LAPV of Volatility Trend Portfolio

Weight strategies at 0.2765, though their final returns were negative, which also shows that staying the same might be a good choice during a market downturn. It is worth noting that PortfolioZero, using the R2 and R3 reward functions, along with SARL, were the only three strategies that managed to profit. This shows that the inclusion of external data played a crucial role, whether for state determination or reward structuring.

Combining these results with the upward trend portfolio analysis, we find that the differences in LAPV among various strategies in a downward trend portfolio are not significant, since the selected experimental dataset generally exhibits a downward trend. Moreover, influenced by the reward functions, PortfolioZero is more sensitive to the market, ensuring overall low risk control and capturing small market rebounds for short-term profits.

4.2.3. Volatility Trend Portfolio

The experimental results of the volatility trend portfolio are shown in Figure 5. In the early backtesting period, the curves of all strategies were nearly identical, showing a slow upward trend. However, starting in mid-May, significant differences began to emerge between the performances of

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different strategies. In particular, during this period, the value of the ANTICOR strategy started to diverge from those of other strategies, while the CORN strategy exhibited a noticeable downward trend. Other strategies, relying on their relatively high return capabilities, can gradually rise. By the end of August, as the overall market declined, the curves of all strategies fell to a lower level. In this dataset, most strategies outperformed the benchmark equal-weighted strategy for the majority of the period. Notably, PortfolioZero_R3 did not exhibit outstanding growth in the volatility trend portfolio; instead, it remained relatively stable. The possible reason is that in a complex and volatile financial market environment, diverse market sentiment significantly influences the decision-making, leading it to adopt more conservative investment strategies.

Table 4: The Backtesting Performance of Different Strategies for the Volatility Trend Portfolio

Strategy	**LAPV**	**SR**	**MDD**	**ARR(%)**	**CR(%)**
Equal Weight	0.0809	0.5731	0.1930	8.4228	43.6508
CORN	-0.5035	-1.9762	0.4657	-39.5596	-84.9444
EG	0.0808	0.5729	0.1930	8.4208	43.6408
ANTICOR	0.1156	0.6559	0.2760	12.2600	44.4183
DDPG_R1	0.1007	0.6524	0.2111	10.5928	50.1737
DDPG_R2	0.1050	0.6838	**0.1780**	11.0676	62.1617
DDPG_R3	0.1083	0.6823	0.2152	11.4338	53.1311
SARL	0.127	0.7547	0.2199	13.5453	61.6027
PortfolioZero_R1	0.1208	0.7172	0.1922	12.8443	66.8408
PortfolioZero_R2	0.1309	0.7867	0.2037	13.9899	**68.6669**
PortfolioZero_R3	**0.1467**	**0.8603**	0.2333	**15.7973**	67.7042

Table 4 presents the performance of the volatility trend portfolio in various evaluation metrics. PortfolioZero_R3 performed best in three key metrics, that is, LAPV at 0.1467, SR at 0.8603, and ARR at 15.7973%. Meanwhile, its CR was only slightly lower than that of PortfolioZero_R2 due to its relatively higher MDD. DDPG_R2 performed best with the MDD metric. It is a pity that although Figure 5 shows ANTICOR performed well in the middle of the year, it did not achieve optimal results in any metrics. We consider the reason is the unstable performance, which, ultimately due to the lack of utilizing additional data information. Overall, the performance of DDPG was inferior to that of PortfolioZero, and even when using the R3 reward func-

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tion, its ARR remained lower than PortfolioZero. SARL, again, performed well but was not better than PortfolioZero_R2 and PortfolioZero_R3.

4.2.4. Performance Analysis

Through backtesting and metric analysis of various types of portfolios, reinforcement learning algorithms have demonstrated distinct performance characteristics in portfolio management. The PortfolioZero excelled in upward, downward, and volatility trend portfolios. On average across the three datasets, PortfolioZero_R3 achieved the best results, with its ARR being 4.9% higher than using SARL and 11.8% than DDPG. This superior performance demonstrates the effectiveness of our model design, which captures deep variations in asset prices more effectively, thereby enabling better decision-making. Additionally, after employing reward functions R2 and R3 based on market sentiment, the performances of both PortfolioZero and DDPG improved noticeably. On average, when PortfolioZero used R3 as the reward function, its ARR increased by 5.5% compared to R1. This not only validates the feasibility of the model in financial market applications but also highlights the influence of market sentiment on investment decisions.

4.2.5. Model Generalization Analysis

While our experiments focused on the Chinese A-share market using ETF data, the PortfolioZero framework is designed to be adaptable to various financial markets and asset types. The architecture of this model, particularly its Transformer-based representation network and sentiment-aware reward mechanisms, can theoretically process any time-series financial data combined with relevant market sentiment information.

For other equity markets (e.g., US, European), our model would remain applicable as these markets share fundamental characteristics with the A-share market, though the sentiment analysis component might need recalibration to account for different market news cycles and investor behavior patterns. However, several considerations would need to be addressed for cross-market application. First, market microstructure differences (e.g., trading hours, tick sizes, transaction costs) would require appropriate adjustments to the model's trading frequency and cost calculations. Second, the sentiment analysis component would need to be retrained with market-specific news sources and potentially modified to capture market-specific sentiment indicators. Third, the reward functions might need reweighting to reflect different risk-return relationships across markets. These adaptations would

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be particularly important for markets with distinct characteristics, such as cryptocurrency markets with 24/7 trading and high volatility.

4.3. Asset Weight Difference Analysis

The equal-weight strategy diversifies risk in portfolio management by distributing asset weights evenly, thereby avoiding over-reliance on specific markets or value-weighting methods. In backtesting, observing significant "peaks" or "troughs" in asset value indicates a marked increase in market volatility, suggesting potential arbitrage opportunities and necessitating stricter risk management. In this section, we selected four stocks out of nine to analyze the significant trough observed on May 16, 2023, and the significant peak on July 25, 2023. These two time points reflect periods of high market volatility. By comparing the weight changes over four trading days before and after these two dates, and conducting a visual analysis, we explored how the PortfolioZero, DDPG and SARL allocate asset under the same extra news.

4.3.1. Market Trough

Figure 6 illustrates the weight allocation strategies of the PortfolioZero and DDPG models of reward function R3 for four stocks in response to market volatility causing a trough, along with the price trends of these assets. Figures 6(a) and 6(d) correspond to stocks that show a downward price trend after experiencing the volatility point. The PortfolioZero model was to maintain the holding weight of that stock at 0 to avoid losses, and SARL held a very low weight and became 0. Conversely, DDPG gradually reduced the weight to 0, resulting in greater losses compared to PortfolioZero during this period. On the other hand, observations from Figures 6(b) and 6(c) reveal that when stock prices were at lower levels, the PortfolioZero model tended to increase its holding proportion of the asset to seek additional future arbitrage opportunities. As stock prices rose, the model correspondingly reduced its holdings to manage potential risks. The performance of SARL was similar but not decisive. In contrast, the DDPG algorithm did not demonstrate this advantage. It is important to note that this is a trough position, so PortfolioZero mostly chose a conservative strategy, trying to reduce stock holdings unless an individual stock showed a particularly strong upward trend. This indicates that it can make decisions by combining macroeconomic characteristics and individual stock features.

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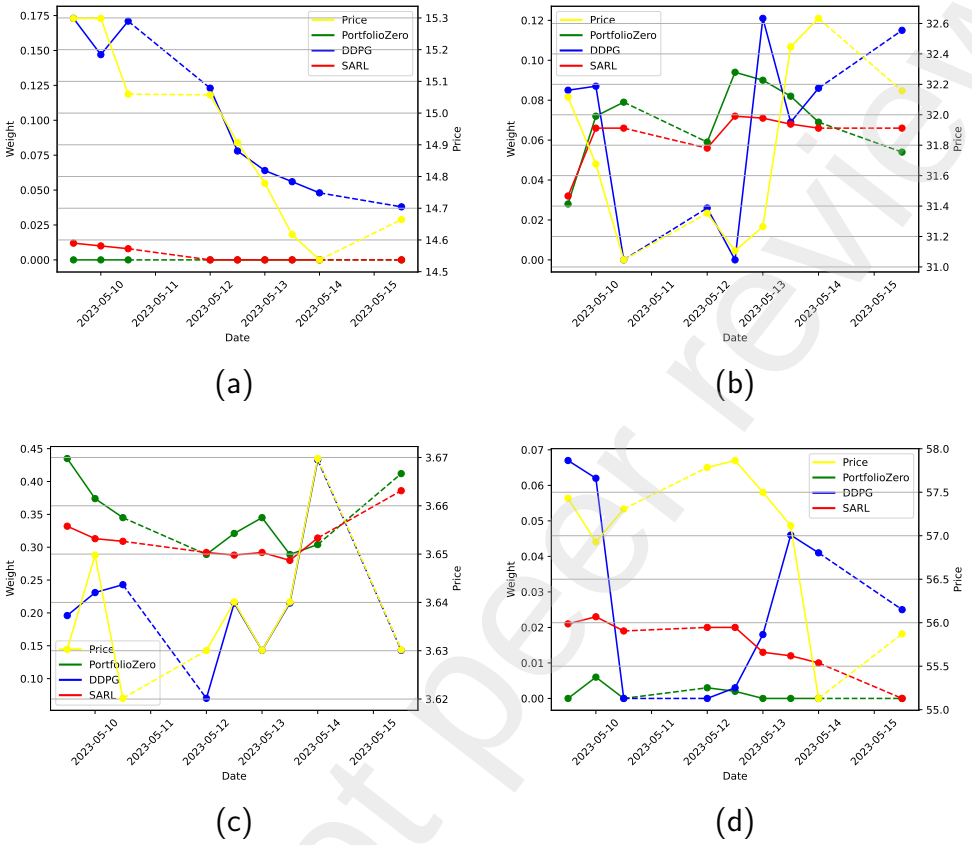


Figure 6: The Change in Stock Weights over the Four Trading Days Around May 16th

4.3.2. Market Peak

Figure 7 displays the allocation strategies for four stocks when the model encounters volatility leading to a market peak. During this period, the stock prices corresponding to Figures 7(b) and 7(c) fell, with a maximum drop of 7.9%. PortfolioZero chose to maintain its holding weight at 0 before this volatility point. Similar to the situation causing market trough volatility, PortfolioZero continued to exhibit a distinct strategy of reducing positions at highs and increasing positions at lows for the other two stocks. This is particularly evident in Figures 7(a) and 7(d); even though the stock prices are high, PortfolioZero still reduced its holdings. Similarly, In contrast, the changes in stock weights output by the DDPG algorithm almost coincided with the trends in stock price changes. Note that in this condition, SARL

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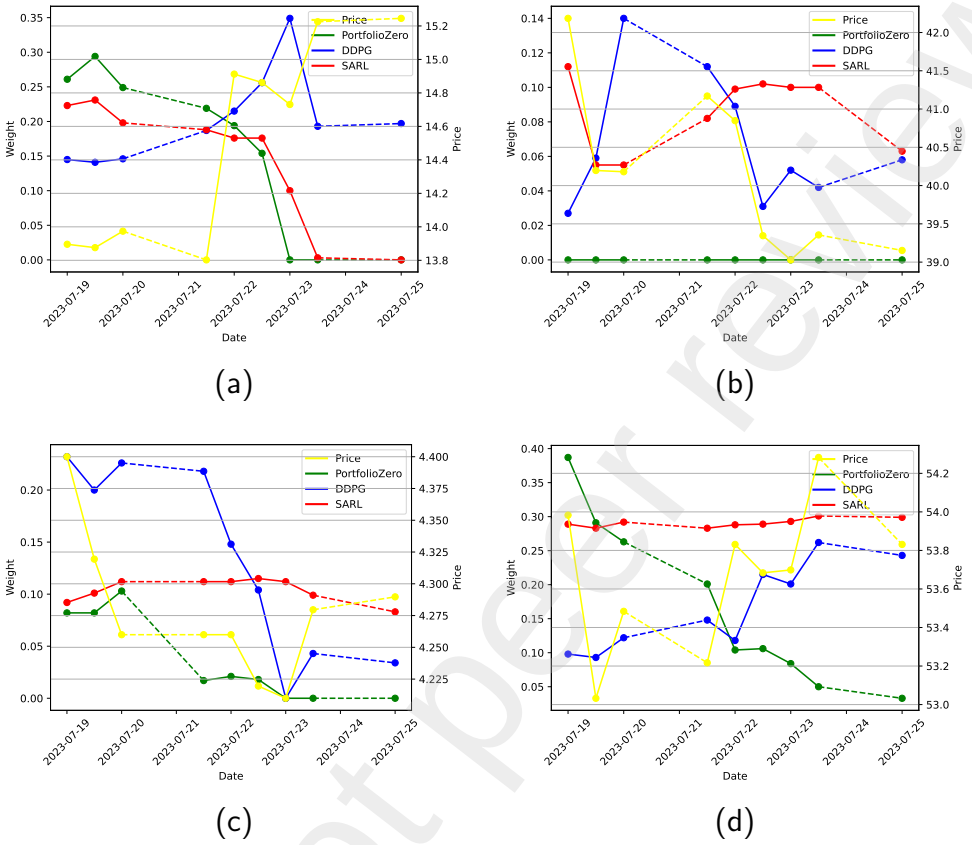


Figure 7: The Change in Stock Weights over the Four Trading Days Around July 25th

did not show the exact same trend as PortfolioZero, especially in Figure 7(b), where they are completely different. This indicates that SARL is somewhat weaker in feature recognition. Additionally, it can be seen that during peak periods, PortfolioZero generally adopts a strategy of reducing holdings due to the pullback after reaching the highest value.

4.3.3. Discussion

By comparing the model performance in the two scenarios mentioned above, we observed the significant differences between the PortfolioZero and DDPG models in asset allocation strategies and risk control measures when facing downward risks. Unlike the low-sensitivity position adjustment approach displayed by the DDPG during this period, changes in PortfolioZero

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were more frequent and decisive. This is because PortfolioZero can leverage its model structure to better sense changes in external information, allowing it to make comprehensive judgments based on macroeconomic trends and micro-level stock prices. Actually, SARL can also detect the risk in most conditions, but its strategy was relatively slower to react to changes.

5. Conclusions

To achieve stable returns through portfolio management remains challenging due to the high randomness and unpredictability of the stock market. In this paper, we conduct research from two aspects. First, we propose a reinforcement learning model named PortfolioZero, which improves the MuZero framework by introducing a Transformer-based pre-trained network to boost the capability representation network, redesigning the model architecture for continuous action spaces, and enabling the model to deeply extract changing trend features from complex asset price data. Second, we incorporate comments on stocks and real-time news data, using the StructBERT model to extract market sentiment features, and design two reward functions that combines sentiment scores with market prices to ensure the sensitivity of model to market changes. We conducted investment simulations using three types of asset portfolios: rising, falling, and volatile. Comparisons were made with traditional portfolio models Equal Weight, CORN, EG, ANTICOR, the reinforcement learning model DDPG, and the state-of-the-art model SARL. On average, across the three datasets, PortfolioZero_R3 outperformed SARL by 4.9% and DDPG by 11.8% in terms of ARR. After employing the reward function R3, the ARR of PortfolioZero can increase by 5.5% compared to directly using the reward function R1.

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Appendix A. The Stock Selection in Experiments

To ensure robustness and minimize potential selection biases, we implemented a systematic selection process based on multiple objective criteria.

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Table A.5: The Stocks Using in Experiments

Type of Portfolio	Stock ID		
Uppward Trend	000063.SZ	000338.SZ	000895.SZ
	000100.SZ	000625.SZ	002050.SZ
	000166.SZ	000725.SZ	002230.SZ
Downward Trend	000001.SZ	000568.SZ	000776.SZ
	000002.SZ	000651.SZ	000858.SZ
	000538.SZ	000661.SZ	002001.SZ
Volatile Trend	000002.SZ	000166.SZ	000568.SZ
	000063.SZ	000338.SZ	000625.SZ
	000100.SZ	000538.SZ	000725.SZ

First, we filtered the Chinese A-share market ETFs using the following quantitative thresholds: 1) Market Capitalization: We selected from the top 50th percentile to ensure adequate market representation; 2) Trading Volume: We minimum average daily trading volume of 1 million shares over the past year to guarantee liquidity; 3) Price History: Required at least 3 years of continuous trading data with no more than 5 consecutive days of trading suspension; 4) Bid-Ask Spread: Maximum average spread of 2% to ensure trading feasibility.

Table A.5 shows all the stocks than were using in our experimental studies.

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